



Arithmetic geometry

In mathematics, **arithmetic geometry** is roughly the application of techniques from algebraic geometry to problems in number theory.^[1] Arithmetic geometry is centered around Diophantine geometry, the study of rational points of algebraic varieties.^{[2][3]}

In more abstract terms, arithmetic geometry can be defined as the study of schemes of finite type over the spectrum of the ring of integers.^[4]

Overview

The classical objects of interest in arithmetic geometry are rational points: sets of solutions of a system of polynomial equations over number fields, finite fields, p-adic fields, or function fields, i.e. fields that are not algebraically closed excluding the real numbers. Rational points can be directly characterized by height functions which measure their arithmetic complexity.^[5]

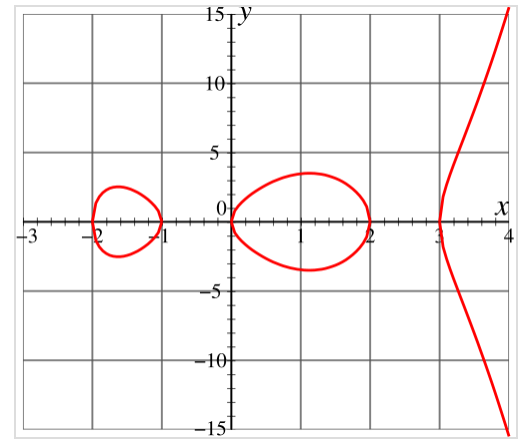
The structure of algebraic varieties defined over non-algebraically closed fields has become a central area of interest that arose with the modern abstract development of algebraic geometry. Over finite fields, étale cohomology provides topological invariants associated to algebraic varieties.^[6] p-adic Hodge theory gives tools to examine when cohomological properties of varieties over the complex numbers extend to those over p-adic fields.^[7]

History

19th century: early arithmetic geometry

In the early 19th century, Carl Friedrich Gauss observed that non-zero integer solutions to homogeneous polynomial equations with rational coefficients exist if non-zero rational solutions exist.^[8]

In the 1850s, Leopold Kronecker formulated the Kronecker–Weber theorem, introduced the theory of divisors, and made numerous other connections between number theory and algebra. He then conjectured his "liebster Jugendtraum" ("dearest dream of youth"), a generalization that was later put



The hyperelliptic curve defined by $y^2 = x(x+1)(x-3)(x+2)(x-2)$ has only finitely many rational points (such as the points $(-2, 0)$ and $(-1, 0)$) by Faltings's theorem.

forward by Hilbert in a modified form as his twelfth problem, which outlines a goal to have number theory operate only with rings that are quotients of polynomial rings over the integers.^[9]

Early-to-mid 20th century: algebraic developments and the Weil conjectures

In the late 1920s, André Weil demonstrated profound connections between algebraic geometry and number theory with his doctoral work leading to the Mordell–Weil theorem which demonstrates that the set of rational points of an abelian variety is a finitely generated abelian group.^[10]

Modern foundations of algebraic geometry were developed based on contemporary commutative algebra, including valuation theory and the theory of ideals by Oscar Zariski and others in the 1930s and 1940s.^[11]

In 1949, André Weil posed the landmark Weil conjectures about the local zeta-functions of algebraic varieties over finite fields.^[12] These conjectures offered a framework between algebraic geometry and number theory that propelled Alexander Grothendieck to recast the foundations making use of sheaf theory (together with Jean-Pierre Serre), and later scheme theory, in the 1950s and 1960s.^[13] Bernard Dwork proved one of the four Weil conjectures (rationality of the local zeta function) in 1960.^[14] Grothendieck developed étale cohomology theory to prove two of the Weil conjectures (together with Michael Artin and Jean-Louis Verdier) by 1965.^{[6][15]} The last of the Weil conjectures (an analogue of the Riemann hypothesis) would be finally proven in 1974 by Pierre Deligne.^[16]

Mid-to-late 20th century: developments in modularity, p-adic methods, and beyond

Between 1956 and 1957, Yutaka Taniyama and Goro Shimura posed the Taniyama–Shimura conjecture (now known as the modularity theorem) relating elliptic curves to modular forms.^{[17][18]} This connection would ultimately lead to the first proof of Fermat's Last Theorem in number theory through algebraic geometry techniques of modularity lifting developed by Andrew Wiles in 1995.^[19]

In the 1960s, Goro Shimura introduced Shimura varieties as generalizations of modular curves.^[20] Since the 1979, Shimura varieties have played a crucial role in the Langlands program as a natural realm of examples for testing conjectures.^[21]

In papers in 1977 and 1978, Barry Mazur proved the torsion conjecture giving a complete list of the possible torsion subgroups of elliptic curves over the rational numbers. Mazur's first proof of this theorem depended upon a complete analysis of the rational points on certain modular curves.^{[22][23]}

In 1996, the proof of the torsion conjecture was extended to all number fields by Loïc Merel.^[24]

In 1983, Gerd Faltings proved the Mordell conjecture, demonstrating that a curve of genus greater than 1 has only finitely many rational points (where the Mordell–Weil theorem only demonstrates finite generation of the set of rational points as opposed to finiteness).^{[25][26]}

In 2001, the proof of the local Langlands conjectures for GL_n was based on the geometry of certain Shimura varieties.^[27]

In the 2010s, Peter Scholze developed perfectoid spaces and new cohomology theories in arithmetic geometry over p-adic fields with application to Galois representations and certain cases of the weight-monodromy conjecture.^{[28][29]}

See also

- Anabelian geometry
- Frobenioid
- Arithmetic dynamics
- Arithmetic of abelian varieties
- Birch and Swinnerton-Dyer conjecture
- Moduli of algebraic curves
- Siegel modular variety
- Siegel's theorem on integral points
- Category theory

References

1. Sutherland, Andrew V. (September 5, 2013). "Introduction to Arithmetic Geometry" (https://ocw.mit.edu/courses/mathematics/18-782-introduction-to-arithmetic-geometry-fall-2013/lecture-notes/MIT18_782F13_lec1.pdf) (PDF). Retrieved 22 March 2019.
2. Klarreich, Erica (June 28, 2016). "Peter Scholze and the Future of Arithmetic Geometry" (<https://www.quantamagazine.org/peter-scholze-and-the-future-of-arithmetic-geometry-20160628/>). Retrieved March 22, 2019.
3. Poonen, Bjorn (2009). "Introduction to Arithmetic Geometry" (<http://math.mit.edu/~poonen/782/782notes.pdf>) (PDF). Retrieved March 22, 2019.
4. Arithmetic geometry (<https://ncatlab.org/nlab/show/arithmetic+geometry>) at the *nLab*
5. Lang, Serge (1997). *Survey of Diophantine Geometry*. Springer-Verlag. pp. 43–67. ISBN 3-540-61223-8. Zbl 0869.11051 (<https://zbmath.org/?format=complete&q=an:0869.11051>).
6. Grothendieck, Alexander (1960). "The cohomology theory of abstract algebraic varieties" (<http://grothendieckcircle.org/>). *Proc. Internat. Congress Math. (Edinburgh, 1958)*. Cambridge University Press. pp. 103–118. MR 0130879 (<https://mathscinet.ams.org/mathscinet-getitem?mr=0130879>).
7. Serre, Jean-Pierre (1967). "Résumé des cours, 1965–66". *Annuaire du Collège de France*. Paris: 49–58.
8. Mordell, Louis J. (1969). *Diophantine Equations*. Academic Press. p. 1. ISBN 978-0125062503.
9. Gowers, Timothy; Barrow-Green, June; Leader, Imre (2008). *The Princeton companion to mathematics* (<https://archive.org/details/princetoncompanion00gowe>). Princeton University Press. pp. 773–774. ISBN 978-0-691-11880-2.
10. A. Weil, *L'arithmétique sur les courbes algébriques*, Acta Math 52, (1929) p. 281-315, reprinted in vol 1 of his collected papers ISBN 0-387-90330-5.

11. Zariski, Oscar (2004) [1935]. Abhyankar, Shreeram S.; Lipman, Joseph; Mumford, David (eds.). *Algebraic surfaces* (<https://books.google.com/books?id=d6Zzhm9eCmgC>). Classics in mathematics (second supplemented ed.). Berlin, New York: Springer-Verlag. ISBN 978-3-540-58658-6. MR 0469915 (<https://mathscinet.ams.org/mathscinet-getitem?mr=0469915>).
12. Weil, André (1949). "Numbers of solutions of equations in finite fields" (<https://doi.org/10.1090%2FS0002-9904-1949-09219-4>). *Bulletin of the American Mathematical Society*. **55** (5): 497–508. doi:10.1090/S0002-9904-1949-09219-4 (<https://doi.org/10.1090%2FS0002-9904-1949-09219-4>). ISSN 0002-9904 (<https://search.worldcat.org/issn/0002-9904>). MR 0029393 (<https://mathscinet.ams.org/mathscinet-getitem?mr=0029393>). Reprinted in *Oeuvres Scientifiques/Collected Papers* by André Weil ISBN 0-387-90330-5
13. Serre, Jean-Pierre (1955). "Faisceaux Algébriques Cohérents". *The Annals of Mathematics*. **61** (2): 197–278. doi:10.2307/1969915 (<https://doi.org/10.2307%2F1969915>). JSTOR 1969915 (<https://www.jstor.org/stable/1969915>).
14. Dwork, Bernard (1960). "On the rationality of the zeta function of an algebraic variety". *American Journal of Mathematics*. **82** (3). American Journal of Mathematics, Vol. 82, No. 3: 631–648. doi:10.2307/2372974 (<https://doi.org/10.2307%2F2372974>). ISSN 0002-9327 (<https://search.worldcat.org/issn/0002-9327>). JSTOR 2372974 (<https://www.jstor.org/stable/2372974>). MR 0140494 (<https://mathscinet.ams.org/mathscinet-getitem?mr=0140494>).
15. Grothendieck, Alexander (1995) [1965]. "Formule de Lefschetz et rationalité des fonctions L" (http://www.numdam.org/item?id=SB_1964-1966__9__41_0). *Séminaire Bourbaki*. Vol. 9. Paris: Société Mathématique de France. pp. 41–55. MR 1608788 (<https://mathscinet.ams.org/mathscinet-getitem?mr=1608788>).
16. Deligne, Pierre (1974). "La conjecture de Weil. I" (http://www.numdam.org/item?id=PMIHES_1974__43__273_0). *Publications Mathématiques de l'IHÉS*. **43** (1): 273–307. doi:10.1007/BF02684373 (<https://doi.org/10.1007%2FBF02684373>). ISSN 1618-1913 (<https://search.worldcat.org/issn/1618-1913>). MR 0340258 (<https://mathscinet.ams.org/mathscinet-getitem?mr=0340258>).
17. Taniyama, Yutaka (1956). "Problem 12". *Sugaku* (in Japanese). **7**: 269.
18. Shimura, Goro (1989). "Yutaka Taniyama and his time. Very personal recollections" (<https://doi.org/10.1112%2Fblms%2F21.2.186>). *The Bulletin of the London Mathematical Society*. **21** (2): 186–196. doi:10.1112/blms/21.2.186 (<https://doi.org/10.1112%2Fblms%2F21.2.186>). ISSN 0024-6093 (<https://search.worldcat.org/issn/0024-6093>). MR 0976064 (<https://mathscinet.ams.org/mathscinet-getitem?mr=0976064>).
19. Wiles, Andrew (1995). "Modular elliptic curves and Fermat's Last Theorem" (<https://web.archive.org/web/20110510062158/http://math.stanford.edu/%7Elekheng/flt/wiles.pdf>) (PDF). *Annals of Mathematics*. **141** (3): 443–551. CiteSeerX 10.1.1.169.9076 (<https://citeseerx.ist.psu.edu/viewdoc/summary?doi=10.1.1.169.9076>). doi:10.2307/2118559 (<https://doi.org/10.2307%2F2118559>). JSTOR 2118559 (<https://www.jstor.org/stable/2118559>). OCLC 37032255 (<https://search.worldcat.org/oclc/37032255>). Archived from the original (<http://math.stanford.edu/~lekheng/flt/wiles.pdf>) (PDF) on 2011-05-10. Retrieved 2019-03-22.
20. Shimura, Goro (2003). *The Collected Works of Goro Shimura*. Springer Nature. ISBN 978-0387954158.
21. Langlands, Robert (1979). "Automorphic Representations, Shimura Varieties, and Motives. Ein Märchen" (<http://www.sunsite.ubc.ca/DigitalMathArchive/Langlands/pdf/autoreps-ps.pdf>) (PDF). In Borel, Armand; Casselman, William (eds.). *Automorphic Forms, Representations, and L-Functions: Symposium in Pure Mathematics*. Vol. XXXIII Part 1. Chelsea Publishing Company. pp. 205–246.
22. Mazur, Barry (1977). "Modular curves and the Eisenstein ideal" (http://www.numdam.org/item/PMIHES_1977__47__33_0). *Publications Mathématiques de l'IHÉS*. **47** (1): 33–186. doi:10.1007/BF02684339 (<https://doi.org/10.1007%2FBF02684339>). MR 0488287 (<https://mathscinet.ams.org/mathscinet-getitem?mr=0488287>).

23. Mazur, Barry (1978). "Rational isogenies of prime degree". *Inventiones Mathematicae*. **44** (2). with appendix by Dorian Goldfeld: 129–162. Bibcode:1978InMat..44..129M (<https://ui.adsabs.harvard.edu/abs/1978InMat..44..129M>). doi:10.1007/BF01390348 (<https://doi.org/10.1007%2FBF01390348>). MR 0482230 (<https://mathscinet.ams.org/mathscinet-getitem?mr=0482230>).
24. Merel, Loïc (1996). "Bornes pour la torsion des courbes elliptiques sur les corps de nombres" [Bounds for the torsion of elliptic curves over number fields]. *Inventiones Mathematicae* (in French). **124** (1): 437–449. Bibcode:1996InMat.124..437M (<https://ui.adsabs.harvard.edu/abs/1996InMat.124..437M>). doi:10.1007/s002220050059 (<https://doi.org/10.1007%2Fs002220050059>). MR 1369424 (<https://mathscinet.ams.org/mathscinet-getitem?mr=1369424>).
25. Faltings, Gerd (1983). "Endlichkeitssätze für abelsche Varietäten über Zahlkörpern" [Finiteness theorems for abelian varieties over number fields]. *Inventiones Mathematicae* (in German). **73** (3): 349–366. Bibcode:1983InMat..73..349F (<https://ui.adsabs.harvard.edu/abs/1983InMat..73..349F>). doi:10.1007/BF01388432 (<https://doi.org/10.1007%2FBF01388432>). MR 0718935 (<https://mathscinet.ams.org/mathscinet-getitem?mr=0718935>).
26. Faltings, Gerd (1984). "Erratum: Endlichkeitssätze für abelsche Varietäten über Zahlkörpern" (<https://doi.org/10.1007%2FBF01388572>). *Inventiones Mathematicae* (in German). **75** (2): 381. doi:10.1007/BF01388572 (<https://doi.org/10.1007%2FBF01388572>). MR 0732554 (<https://mathscinet.ams.org/mathscinet-getitem?mr=0732554>).
27. Harris, Michael; Taylor, Richard (2001). *The geometry and cohomology of some simple Shimura varieties* (<https://books.google.com/books?id=sigBbO69hvMC>). Annals of Mathematics Studies. Vol. 151. Princeton University Press. ISBN 978-0-691-09090-0. MR 1876802 (<https://mathscinet.ams.org/mathscinet-getitem?mr=1876802>).
28. "Fields Medals 2018" (<https://www.mathunion.org/imu-awards/fields-medal/fields-medals-2018>). International Mathematical Union. Retrieved 2 August 2018.
29. Scholze, Peter. "Perfectoid spaces: A survey" (http://www.math.uni-bonn.de/people/scholze/CD_M.pdf) (PDF). *University of Bonn*. Retrieved 4 November 2018.

Retrieved from "https://en.wikipedia.org/w/index.php?title=Arithmetic_geometry&oldid=1321395466"